

Memory Assignment

Memory Allocation Project using Segmentation

Segmentation is a memory-management scheme that supports user view of a program. A program is a collection of segments.

Project objectives:

Implementation of Allocation and deallocation techniques using segmentation.

Data Structure used:

a) allocated partitions table b) free partitions (holes) table

You should update these tables after each allocation or deallocation.

Since segments vary in length, memory allocation is a dynamic storage-allocation problem. You should implement both algorithms: **First-Fit and Best-Fit allocation algorithms.**

Inputs:

- 1- User inputs total memory size
- 2- User inputs holes starting address and size.
- 3- User inputs Processes one by one.
- 4- User inputs for each process:
 - a. Number of segments
 - b. Name and size of each segment

(Example: P1 has 3 segments ... Code: 50, Data: 200, Stack:100)

- 5- User inputs the method of allocation (first fit or best fit).

Scenario to be done:

- 1- Allocate segments using either First-Fit or Best-Fit allocation methodology.
- 2- If one segment or more of a process can not fit in any hole you should generate a message to state that this process does not fit.
- 3- Deallocate a process (The user chooses a process to deallocate, you should deallocate all segments of this process and consider their spaces as holes to be used later and add them to any neighbouring holes).

Output:

Your output is a drawing representing memory layout after each allocation or deallocation. Also, you should show the contents of the segment table for each process.

Deliverable: On LMS deliver a zipped file including a pdf report with screenshots of your project + link to exe file uploaded on drive + code files